

DATA DOCUMENTATION

Nigeria 2026 Federal Government Appropriation Bill Analysis

SECTORS IN CHECK: Education · Health · Security

Team Name: Data Guardian (Group 22)

Documentation	Detail
Document Purpose	Team record of every sourcing, cleaning, and structuring decision made before analysis
Sectors Covered	Education, Health, Security (For the Federal Government of Nigeria, 2026 fiscal year)
Primary Source	Budget Office of the Federation (BOF)
Source Format	Official PDF - 2026 FGN Appropriation Bill
Extracted Datasets	4 excel workbooks produced by the team from the source PDF
Date of Access	May 2026

1. Data Sourcing

1.1 Where the Data Came From

All figures used in this analysis originate from one primary source.

Attribute	Detail
Source Name	Budget Office of the Federation (BOF)
Parent Institution	Federal Ministry of Finance, Budget and National Planning — Federal Republic of Nigeria
Document	2026 Federal Government of Nigeria Appropriation Bill
Original Format	PDF

Extracted Files	1. EDUCATION.xlsx 2. SECURITY.xlsx 3. HEALTH.xlsx 4_FEDERAL_BUDGET(SECTORS-MDA).xlsx
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1.2 Why the Team Trusted This Source

The Budget Office of the Federation is the statutory body mandated to prepare and publish the federal budget. Its appropriation documents are:

- The official, legally binding record of government expenditure as presented to the National Assembly.
- The same documents referenced by MDAs, auditors, and oversight bodies, not a third-party compilation.
- Consistent in MDA coding and structure with prior-year appropriation bills, confirming authenticity.
- Free of commercial or editorial influence, the figures are government-generated at source.

No external dataset, news report, or secondary publication was used. The BOF PDF is the sole numerical source for this project.

2. Data Extraction

2.1 What Was in the Original PDF

The Appropriation Bill PDF contained:

- A national MDA summary table: total allocations across all 56 government sectors.
- Funds summary allocation for main envelop – personnel, main envelop – overhead, crf charges - pensions and gratuities, crf charges - service wide votes, crf charges - capital supplementation, statutory transfers, debt services, capital development fund main.
- Agency-level breakdowns for every MDA, with personnel, overhead, and capital figures per agency.
- Detailed line items within each agency, showing exactly what each naira allocation covers.
- Individual capital projects listed by name, ERGP code, allocation, and status (Ongoing / New).

2.2 What the Team Extracted

Extraction was manual and deliberate. The team pulled only what was needed for the three-sector analysis, with the use of AI tools, Excel and Python.

Workbook	Content Extracted	Sheets Built
FEDERAL BUDGET(SECTORS-MDA).xlsx	National MDA summary; all 56 sectors, used for share and comparison analysis	1 sheet
EDUCATION.xlsx	Full Education sector data: MDA summary, all 268 agencies, 15,438 line items, 6,758 projects	4 sheets

SECURITY.xlsx	Full Security sector data: 5 MDAs summary, 43 agencies, 2,057 line items, 793 projects	4 sheets
HEALTH.xlsx	Full Health sector data: MDA summary, 149 agencies, 5,458 line items, 3,989 projects	4 sheets

2.3 Structure of Each Sector Workbook

Every sector workbook was built with the same four-layer hierarchy, allowing analysis to move from the widest view down to individual projects without losing context:

Sheet	Contents	Rows (approx.)
MDA SUMMARY	Top-level MDA summary- total personnel, overhead, capital, and grand total for the entire sector	1–5
AGENCY SUMMARY	One row per agency under the MDA, each with its own personnel / overhead / capital / total breakdown	43–268
LINE ITEMS	Every expenditure line item within each agency, tagged by budget category (Personnel / Overhead / Capital)	2,057–15,438
PROJECTS	Individual named capital projects with ERGP code, allocation, and status (Ongoing or New)	793–6,758

3. Cleaning Decisions

3.1 What Was Messy

Issue	Location	Impact
PDF is not machine-readable — all figures required manual extraction into structured tables	Entire Appropriation Bill PDF	High
Numeric columns in line-item sheets contain NaN/blank cells, each row only populates the column relevant to its budget category	LINE ITEMS across all three sector files	Medium
Minor column naming inconsistency across workbooks (e.g. 'personnel_naira' in Education, Health and Security to 'total_personnel' in Health, Education and Security)	Agency Summary and Line Item column headers	Low
Education sheet named 'LINE ITEM' (singular); Health and Security named 'LINE ITEMS' (plural)	Sheet tab names	Low
Line Items and Projects were together in the same sheet giving errors due to the	Line Items Sheet in SECURITY.xlsx, HEALTH.xlsx AND EDUCATION.xlsx	High

difference in code characters (Numerical and alphanumerical codes)		
Security sector spans five separate MDAs (Defence, Interior, Police Affairs, National Security Adviser, Police Service Commission) rather than a single MDA	SECURITY.xlsx - MDA SUMMARY sheet	Low

3.2 What Was Fixed

- NaN cells in line-item numeric columns: retained as it was documented. Each line item belongs to exactly one budget category, so blank values in the other two cost columns are structurally correct, not errors. They were not filled with zeros to avoid misrepresenting the data.
- Column naming inconsistency: noted and mapped in the analysis scripts. Columns refer to the same concept (personnel cost, overhead cost, capital cost) across all files; the naming difference does not affect values.
- Education 'LINE ITEM' tab: renamed to 'LINE ITEMS' in analysis references for consistency. The underlying data was unchanged.
- Line Items and Projects separated into two separate sheets for accuracy and clear interpretation.
- Security multi-MDA structure: the MDA SUMMARY sheet lists each of the five MDAs as a separate row. Analysis treats these as sub-components of the Security sector and sums them where a single sector total is needed.

3.3 What Was Dropped and Why

Dropped Element	Reason
All sectors outside Education, Health, and Security (from the national summary)	Out of scope, retained only as denominators for sector share calculations
Retained Revenue column (zero-valued across all three sector files)	No data present; column would add no analytical value
Aid/Grant Funded column (zero-valued across all three sector files)	Same reason as above, all values were zero
Surplus/Deficit reconciliation figures from the source PDF	These are comparative accounting entries against prior-year figures, not budget allocations. Kept in raw notes; excluded from working dataset to prevent misrepresentation
Constituency/ERGP sub-line items already captured in the Projects sheet	Extracting them again from the Line Items sheet would double-count capital expenditure

4. Final Dataset Structure

4.1 National Sector Summary

File: FEDERAL BUDGET(SECTORS-MDA).xlsx ; Sheet name: SECTORS BUDGET ; 56 rows (one per MDA)
; Sheet name: SUMMARY BY FUNDS ; 8 rows

Column	Description	Type
serial_no	Sequential row number	Integer
code	MDA code assigned by the Budget Office of the Federation	Integer
name	Full official MDA name	Text
total_personnel	Aggregate personnel cost for the MDA (₦)	Numeric
total_overhead	Aggregate overhead cost for the MDA (₦)	Numeric
total_capital	Aggregate capital expenditure for the MDA (₦)	Numeric
total_allocation	Grand total allocation: personnel + overhead + capital (₦)	Numeric

4.2 Sector Workbook Schemas

All three sector workbooks share the same four-sheet structure. Column names are consistent within each sheet type; minor naming differences across workbooks are documented in 3.2

MDA SUMMARY — columns: `mda_code` · `mda_name` · `total_personnel` · `total_overhead` · `total_capital` · `total_allocation`

AGENCY SUMMARY — columns: `agency_code` · `agency_name` · `total_personnel` · `total_overhead` · `total_capital` · `total_allocation`

LINE ITEMS — columns: `agency_code` · `agency_name` · `item_code` · `item_description` · `budget_category` · `total_personnel` · `total_overhead` · `total_capital` · `total_allocation`

PROJECTS — columns: `agency_code` · `agency_name` · `item_code` / `project_code` · `item_description` / `project_description` · `total_allocation` · `project_status` (ONGOING or NEW)

4.3 Dataset Readiness Checklist

Check	Status
Source figures cross-checked against original BOF PDF	✓ Confirmed
All four workbooks open without errors and pass row-count validation	✓ Confirmed
No duplicate agency rows within any workbook	✓ Confirmed

Zero-value columns excluded from primary working dataset	✓ Retained Revenue and Aid/Grant columns removed
NaN cells in line items documented and treated as structural (not errors)	✓ Documented in 3.2
Column naming inconsistencies noted and mapped	✓ Documented in 3.2
Security multi-MDA structure accounted for in aggregation logic	✓ Confirmed — 5 MDAs summed for sector total
No external data introduced at any stage	✓ BOF PDF is the sole source of all figures

5. Team Integrity Statement

This record documents every material decision the team made, from downloading the PDF to handling the dataset to analysis. No figures have been invented, assumed, or adjusted to fit a narrative.

Where the data contained structural flukes (NaN cells, naming inconsistencies, multi-MDA security sector), those quirks are described here with the reasoning behind each decision. Where figures were excluded, the exclusion is explained.

The Budget Office of the Federation is the sole source of all numerical data in this project.